Advanced Computing Project (COM7014)

Module Handbook

MSc Data Science

MSc Cyber Security

Level 7

Contents

Section A: Module Overview

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Level:	**7**
Credits:	**60**
Learning Hours:	**600**

Aims

The Advanced Computing Project is a key step in achieving your master's degree. It provides you with the opportunity to address a computing problem through the design, building, testing and evaluation of an artefact. In addition, you will develop your ability to make critical and evaluative judgements in the context of your chosen topic area.

Building on the taught elements of the programme, you will undertake appropriate background research and data gathering in an area relevant to your qualification theme to inform the creation of an appropriate computing artefact.

The module is initially supported by workshop sessions that are designed to consolidate all the previous project management and technical skills that have been conveyed during the taught elements of the programme.

You will be allocated to a supervisor within your chosen topic area to support you through the project process and in the development of your project proposal.

By completing the advanced computing project, you will be able to demonstrate that you are able to produce practical outputs that are relevant to industry and possess the requisite skills to manage a project successfully.

To achieve the module learning outcomes, you are required to produce:

- **A project proposal**
  (no mark attached, but this is a requirement to progress to the project stage)

- **A project portfolio (consisting of a report and software artefact)**
  12,000 words (or equivalent) - worth 100% of the final overall grade.

Module Learning Outcomes

On successful completion of this module, you will be able to:

1. Select, evaluate, and apply critical thinking to an organisational issue or problem within the Computing field.
2. Review literature and methodologies to design and carry out appropriate research and development activities.
3. Compare and contrast the relative merits of different technical development processes and their relevance in different situations.
4. Identify and troubleshoot issues in a logical and critical manner.
5. Work independently, taking responsibility for the project process and where necessary the self-learning of new skills as part of your continuing professional development.
6. Demonstrate effective planning, and project management to develop and present an appropriate computing artefact.
7. Demonstrate an awareness of ethical issues, sustainability, social responsibility, and the potential global context of your chosen project topic.

Graduate attribute

8. Critically evaluate and further contextualise understanding of the subject area with the ability to link the discipline(s) to local, national, and global issues, research and pursue evidence-based arguments within the discipline using an extensive range of academic and professional body resources. Show clear ability to engage with different traditions of thought, and the ability to apply their knowledge in practice including in multidisciplinary or multi-professional contexts.

Module Delivery

Depending on the mode of study, the module will be delivered either through on-campus or online workshops that focus on concepts, theories, and processes. You will also receive individual support from your designated supervisor (once your project title has been approved), who will agree a schedule with you and agree landmarks and objectives. This is intended to motivate you and ensure that you can monitor your own progress.

Within this structure the emphasis will be on applying knowledge in practical tasks and ensuring that you receive ongoing feedback. You will be expected to attend all scheduled workshops and take a proactive approach to your work.

You will be undertaking a significant Computing related project (on a topic of your choice) consisting of two main phases. As well as showcasing your technical skills, this also allows you to demonstrate your ability to plan, implement and deliver a successful project.

Phase 1: Project Proposal & Ethical Approval

This is a gatekeeper document i.e. The proposal does not carry any weighting at this stage, but the content will support your main project report. Your initial proposal **must** be officially approved by your supervisor and module lead (and in some cases the Arden Research Ethics Committee) before you can proceed with your project.

As part of the proposal, you must consider the Arden ethical guidelines and apply for ethical approval (which **must** be in place before any data collection begins). Once ethical permission has been approved and signed off, you can proceed with Phase 2 of the project. It is important to factor in time for the ethics approval process as part of your project plan.

Note: you will *not* be able to submit your final project report to the portal without having completed this phase. In the event of not gaining ethical approval, you will be required to re-design and resubmit your plan based on the feedback and recommendations from your supervisor.

Phase 2: The Main Project Report
The Project report with a final deliverable due in your final semester and submitted via the portal on iLearn. This accounts for **100%** of the total mark.

Section B: Project Selection & Supervision

You will first need to identify and select a suitable project. The final project proposal **must** be agreed by your supervisor, the module leader (and possibly the Research Ethics Committee) before you can begin any data collection. You should check that your project idea meets the following criteria:

- It can be completed within an appropriate time frame.
- Access to any specific hardware/software/specialist resources has been considered.
- It is sufficiently challenging (but not impossible to complete).
- Is original in nature (e.g., not just re-using existing assessment submissions from previous modules or existing work)
- It satisfies the ethical requirements as set out by Arden University (see the Arden Ethics Policy (QA 39 - AU Ethics Policy v7.pdf (arden.ac.uk).
- It is experimental or developmental (i.e. not just an extended literature review)

Project Supervision

Once you have registered and submitted an appropriate project working title through the Arden Ethics Portal (https://ethics.arden.ac.uk/) you will be allocated a project supervisor who will provide you with support throughout the duration of your project. This will include:

- Monitoring and ensuring your progress towards your objectives
- Providing academic guidance and answering general project queries
- Encouraging you to keep appropriate records of actions/progress, such as reference sources, designs, and contact logs
- Providing guided feedback on your project elements (proposal and final draft)

A successful project will require significant thought and preparation. This is not a 'taught' module where there is delivery and assessment of a fixed body of knowledge, or where there is a "model answer". This is a module that is yours, and for you alone, to progress and be responsible for, with advice from the project team and your supervisor, who will act as a guide and facilitator throughout the project process.

Your supervisor will provide you with feedback as to the project's viability and discuss with you any amendments that need to be made before you can proceed. **For this to happen you should ensure that you have contacted your supervisor within the <u>FIRST</u> week of being allocated** (your Module Leader will email you to confirm your supervisor allocation).

Your supervisor will discuss specific details of how the working relationship will be implemented in practice. This is an ideal time to raise any concerns about the development of your initial proposal and how the project will proceed.

To make the most effective use of your supervisor's time you should come prepared for meetings. You should have a list of key points that you wish to discuss and keep notes of any important decisions made or outcomes. Following the meeting, update the project log on ARMS with a summary of the key points that were addressed and any decisions that were agreed.

Section C: Project Structure

Your project will consist of THREE parts: a project proposal, a project report, and your project artefact.

Project Proposal (Submitted via ARMS)

The project proposal is not awarded a grade, but will need to address the following areas before approval can be granted:

- **Overview/Rationale** - an overview of your project and the reason for undertaking this e.g., which issue, or problem is it attempting to solve?
- **Aim** - the aim is a statement of intent that should state what you plan to achieve with the project overall (but not *how* you will do it)
- **Project Objectives** - the objectives define *how* you are going to achieve your aim. Objectives should be **S**pecific, **M**easurable, **A**chievable, **R**esourced and **T**ime-limited (SMART). Information on developing SMART objectives can be found on iLearn.
- **Project Methodologies –** identify suitable development and research methodologies for your project. Provide a clear rationale and justification for your choice. Also include details of:
    - Any data collection that will be required and the methods you intend to use
    - The recruitment of participants for your project

- **Data Management** – describe the data you expect to acquire or generate during the project, how you will manage, analyse, and store the data securely and what mechanisms you will use to share and preserve your data in accordance with Arden regulations.

- **Project Plan –** propose a draft plan of work for the project. All necessary project steps and tasks should be identified along with a realistic timescale. This should be presented as a GANTT chart and uploaded as an attachment with your proposal on ARMS.

Main Project Report (Uploaded via iLearn - 12,000 Words or equivalent)

The project report should include the following sections:

- **Abstract** - A short summary of your project that covers the key areas and findings from your work.

- **Project Framing** – This should cover the background to your project, details of the chosen topic area/problem being addressed by the project, discussion of the rationale, plus the project aim and objectives.

- **Fact Finding** - This section should include details about aspects of the problem that will set the context for the project plus a requirements analysis. This section should be supported by good quality academic and technical sources and data.

- **Project Development** - This section should address the scope, design and implementation of the artefact being developed, as well as the analysis and development work and background research behind this. This will include evidence of your skills in critical analysis, technical design, and research, as well as the choice and the application of appropriate development methodologies.

- **Critical Evaluation** - This section should evaluate the project deliverables and project process. This should address the strengths and weaknesses of the completed artefact, methodologies or other work carried out, the answers to the research questions, and what went well or badly in the project or might have been done differently.

- **Conclusion** – A summary of the main findings from your work as well as suggestions/ recommendations for further development and areas of possible further investigation or entrepreneurial potential.

- **References & Appendices –** Include references for all sources that you have used during your project. The appendices section should be used for supporting material and project evidence unsuitable for the main body of the report (e.g., large datasets or sections of code).

Formative Feedback

To make sure you are on the right track throughout the writing and development process you will share chapters with your supervisor allowing for formative feedback opportunities, which will help the final submission to take shape (**one** review per chapter, plus **one overall** review of the final draft).

You will also have opportunities to discuss your progress in developing your project artefact. You should ensure that you fully engage with the opportunities provided for feedback and communicate regularly with your supervisor, providing chapters for review on a regular basis.

Marking Scheme		**0 – 29%**	**30 – 39%**	**40 – 49%**	**50 – 59%**	**60 – 69%**	**70 – 79%**	**80%+**
Element ↓	**Weighting** ↓							
Project Framing *- Aim/rationale* *- Objectives*	**20%**	No aim and no rationale. Objectives have not been met.	Weak aim and rationale lacking relevance. Basic objectives have been partially met.	Basic aim and rationale, but these lack either clarity, relevance, or depth. Suitable objectives, although these may not have all been achieved.	A clear aim and rationale, but with scope for more development in relevance or depth. Suitable objectives with the majority of these being achieved.	Aim and rationale are clearly outlined, but minor improvements possible in relevance or depth. Suitable objectives which have all been achieved.	Clear aim and rationale with justification. Only minor improvements possible in one area. Challenging objectives that have all been met.	Exemplary aim and rationale, clear and relevant. Challenging objectives which have all been met and exceeded in some cases.
Fact Finding *- Quality of supporting sources* *-Requirements analysis*	**20%**	No supporting sources, or unsuitable sources used. No identification or discussion of potential requirements.	Little use of quality sources, or very little relevance to the project. Little identification or discussion of potential requirements.	Some relevant sources used of an acceptable quality, but these may lack depth and/or relevance to the project. Some identification and discussion of potential requirements but this is limited.	Mostly appropriate sources used but there may be a lack of clarity and or relevance in some parts. Discussion of requirements present but there may be some gaps.	Well selected sources that support the project. There may be some gaps in presentation, clarity of quality. Discussion of almost all requirements, presented clearly, with only minor flaws.	All sources support the project. There may be minor issues with presentation, clarity, or quality. Detailed discussion of all requirements, with only minor flaws.	Comprehensive use of supporting sources of the highest quality. All requirements identified clearly and professionally presented.
Project Development and Implementation *- Quality of Artefact* *- Critical Reflection*	**30%**	Basic artefact that is non-functional or extremely limited – not even at a prototype stage. No critical reflection on use of design methodologies to inform development.	Simple artefact produced but has major flaws and very limited functionality. Very limited critical reflection of design methodologies.	An artefact has been produced but this has functionality issues and lacks technical complexity or has serious flaws. Some critical reflection of design methodologies, but mostly superficial.	A competent artefact has been produced but there may be technical issues preventing full functionality and some flaws in execution. Clear critical reflection of design methodologies, with only some areas lacking depth.	An artefact has been produced that is almost fully functional, with only minor issues. Comprehensive critical reflection of design methodologies, with only minor improvements possible.	A suitable artefact has been produced that is fully functional and demonstrates high technical competence. Deep critical reflection of design methodologies, with only minor improvements possible.	An exceptional and sophisticated artefact produced to a commercial standard. Meets all functional, non-functional and security requirements. Deep reflection on how the artefact has been informed by appropriate supporting data.

Critical Evaluation	**20%**	No evaluation present, or incomplete. No suggestions for future work or development.	Some attempt at a critical evaluation, but this may be either incomplete, descriptive, or lacking in sufficient depth.	A critical evaluation has been carried out, but this may lack depth in many parts or not comprehensively cover all the relevant points.	A critical evaluation has been carried out, but this may lack depth in some parts or not comprehensively cover some relevant points.	A clear critical evaluation has been carried out, but this may lack depth in one area or lack fluency.	A high standard critical evaluation with little room for improvement. There may be minor improvements possible in terms of depth or scope.	An outstanding in-depth critical evaluation.
Documentation & Quality *- Structure* *- Reference Quality* *- Supporting Documentation (appendices)*	**10%**	Contains errors throughout, making most of the work difficult to follow. References are absent, unsuitable, or incomplete, with no adherence to AU Harvard standards. Little or no documentation relevant to the project is presented.	Some structure is evident, but contains errors throughout, making at least half of the report difficult to follow. Most references are absent or incomplete, with little adherence to AU Harvard standards. Majority of sources dated or irrelevant. Only a small amount of relevant documentation has been presented which does not cover the whole project.	A basic structure is evident, but contains errors in some parts, making many sections of the report difficult to follow. Many references are absent or incomplete, but some adherence to AU Harvard standards has been shown. Many sources dated or irrelevant. Evidence of some relevant documentation, but there may still be large gaps or does not relate to the whole project.	A workable structure that meets the basic requirements of a project report but has areas for improvement in several areas. Most references are correctly cited, but not fully in compliance with AU Harvard standards. Not all sources are high quality or relevant. Clear documentation that covers most of the project but may be missing details in a few areas.	A suitable structure that communicates most of the project work. Issues in grammar/presentation may be present in some areas. Almost fully referenced in accordance with AU Harvard standards. Minor improvements possible in citations and selection of reference sources. Well presented project documentation but missing detail in one area.	A near professional standard of communication is evident, with only minor areas of improvement possible. Fully referenced in accordance with AU Harvard standards. Minor improvements possible in selection of reference sources. Comprehensive project documentation with only minor areas for improvement or clarity.	A thoroughly professional standard of communication is evident throughout the work. Fully referenced in accordance with AU Harvard standards with an outstanding selection of relevant reference sources. Project is worthy of publication and further dissemination. Complete, professional project documentation included covering all phases of the work.

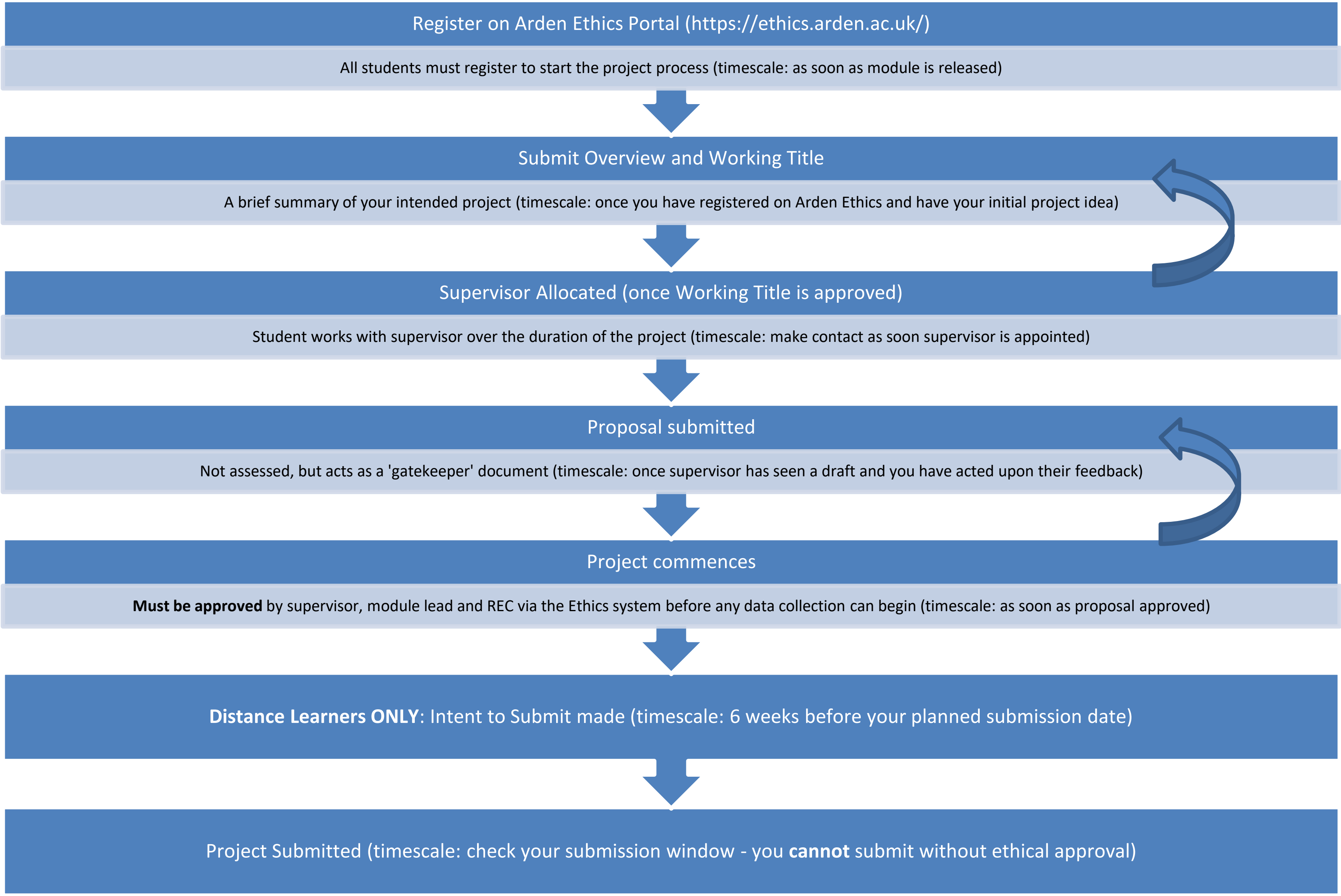

Figure 1 - Computing Project Process

Section D: Submission

It is vital that your supervisor is allowed sufficient time for the reading of drafts, as they also supervise other students and may teach on several modules. You MUST NOT EXPECT your supervisor to read work or respond instantly when a deadline is approaching. Ideally, a full draft of the project report should be sent to your supervisor no later than a **month** before your submission deadline to allow time for changes and amendments.

Submission Arrangements:

When you are ready to submit your project for marking, you will be required to upload an electronic copy of your project report to the portal on iLearn. There is no requirement to submit a hard copy of your work. **You must have ethical approval in order to access the portal**.

You may upload your artefact to the School of Computing Web Server at the end of your project.

Submission Format

Your project should be presented professionally, in the following format, following the suggested structure outlined in section C:

- A4 layout using double line spacing. The recommended font is Arial, size 12 for the main content and depreciating font sizes for sub-headers and headers etc.
- All text should be justified, so that it is straight edged (like a book).
- Any pages preceding those of the main text should be numbered at the centre of the foot of each page.
- Make sure that your project report reads well. Keep paragraphs short and use appropriate headings. Pay particular attention to grammar and sentence construction. Keep content clear, to the point and jargon free.
- Figures and Tables should be clearly labelled, referenced sequentially as they appear in the text and produced via software packages. These should either be placed on a separate page or within the text (but as close to the text at which it is referred to). Where appropriate, acknowledgement of the source should be presented on the page beneath the Figure/Table.
- Make sure that you include an automated Contents Page using appropriate and correct numbering.

Section E: Recommended Reading & Resources

Key Text

Dawson, C.W. (2015) *Projects in computing and information systems: a student's guide.* Third edition. Pearson. Available at: https://search.ebscohost.com/login.aspx?direct=true&AuthType=sso&db=cat08629a&AN=alc.EDZ0002398620&site=eds-live

Other:

Brewer, J & Dittman, K. (2023) Methods of IT Project Management, Fourth Edition. West Lafayette, Indiana: Purdue University Press. Available at: https://research.ebsco.com/linkprocessor/plink?id=4e50fa7b-2431-3278-abb9-7611a890e847

Montasari, R & Jahankhani, H (2021) *Artificial Intelligence in Cyber Security: Impact and Implications: Security Challenges, Technical and Ethical Issues, Forensic Investigative Challenges*. Cham: Springer. Available at: https://research.ebsco.com/linkprocessor/plink?id=722bf0e6-0585-3eea-a4e7-2956bf5c75d7

Wysocki, R. (2019*) Effective Project Management: Traditional, Agile, Extreme, Hybrid*. Indianapolis, IN: Wiley. Available at: https://search.ebscohost.com/login.aspx?direct=true&AuthType=sso&db=nlebk&AN=2099564&site=eds-live

Academic journals:

The British Computer Society - http://www1.bcs.org.uk/
The Institution of Electrical Engineers (IEE) - http://www.iee.org.uk
The Institution of Engineering and Technology - https://ietresearch.onlinelibrary.wiley.com/
The Institute of Electrical and Electronics Engineers (IEEE) Computer Society - http://www.computer.org
The Association of Computing Machinery (ACM) - http://www.acm.org

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